Platform Overview

This platform addresses urban mobility challenges using machine and deep learning models for estimated time of arrival prediction, traffic flow prediction, and parking availability. It leverages synthetic traffic data generated via SUMO simulations to showcase live model performance. The platform also simulates concept drift, triggering model retraining or adaptation to maintain accuracy. An intuitive user interface offers detailed metrics and visualizations, making it a practical and insightful tool for exploring model effectiveness in dynamic urban environments.

Platform Description

The platform is designed to address and showcase solutions for urban transportation challenges, using machine and deep learning tools and techniques. It features pre-trained, optimal models for three key urban mobility problems: estimated time of arrival prediction, traffic flow prediction, and parking availability. It seamlessly integrates live demonstrations of these models, providing users with an interactive and practical approach to solving urban mobility problems.

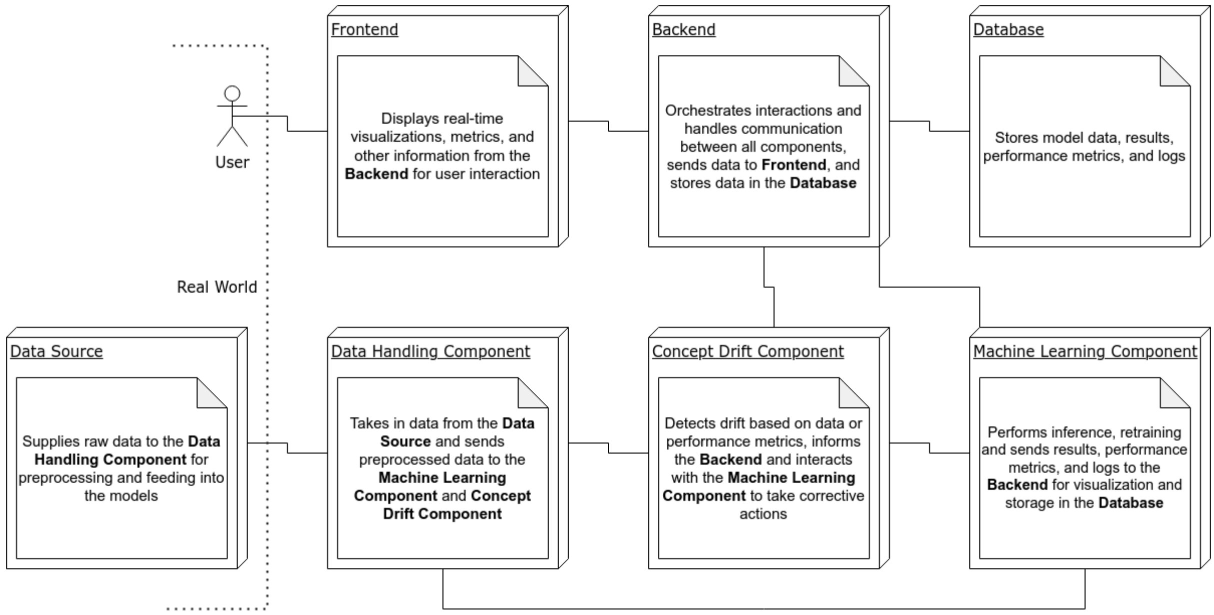
The models are tested in real-time through an integrated SUMO (Simulation of Urban Mobility) simulation, offering a dynamic and interactive visualization of predictions and accuracy metrics. Users can observe live model performance and understand how the models handle real-time data and evolving urban scenarios.

The user interface is designed to be intuitive and user-friendly. The platform continuously tracks model accuracy, offering insights into how each model performs under changing conditions. Detailed performance metrics and visualization tools are provided to help users evaluate and compare the effectiveness of different models and techniques.

A unique aspect of the platform is its ability to simulate concept drift by introducing significant changes in the simulation, such as increased traffic or changes in weather conditions, in order to degrade model performance. Concept drift detection mechanisms then trigger retraining or the use of other adaptation techniques to maintain accuracy in changing conditions.

In summary, this platform is an innovative and comprehensive tool that showcases advanced models in a live, interactive environment. This makes it an invaluable resource for demonstrating cutting-edge technology and providing live and practical visualizations of model performance.

Platform Architecture



Data Source

- **Purpose**: Feeds live or pre-recorded data into the system. It will simulate traffic data, which could also include concept drift like weather changes or traffic surges.
- **Key Functions**:
  - Simulate/stream data from SUMO or recorded sources.
  - Trigger concept drift conditions based on rules or schedules.
  - Provide data to other components (e.g., backend, concept drift detector).
- **Interaction**:
  - Supplies raw data to the **Data Handling Component** for preprocessing and feeding into the models.

Data Handling Component

- **Purpose**: Acts as an intermediary between the Data Source and other system components like the ML component. Responsible for data preparation, transformation, and dispatching.
- **Key Functions**:
  - Preprocess incoming data (normalize, format) for model inference.
  - Feed data to the Machine Learning Component.
  - Send relevant data to the Concept Drift Component for analysis.
- **Interaction**:
  - Takes in data from the **Data Source** and sends preprocessed data to the **Machine Learning Component** and **Concept Drift Component**.

Concept Drift Component

- **Purpose**: Monitors data or model performance to detect concept drift and initiate adaptation measures.

- **Key Functions**:
  - Perform drift detection on incoming data or on the model's performance metrics.
  - Trigger drift adaptation (e.g., retraining, switching to backup models, or alerting the user).
  - Interact with the ML Component for retraining or adaptation.
- **Interaction**:
  - Detects drift based on data or performance metrics, informs the **Backend** and interacts with the **Machine Learning Component** to take corrective actions.

Machine Learning Component

- **Purpose**: Hosts the pre-trained models and performs the core machine learning tasks, including inference, retraining, and evaluation.
- **Key Functions**:
  - Load and maintain the pre-trained models.
  - Perform inference based on incoming data.
  - Handle retraining or switching to new classifiers when drift is detected.
  - Log model performance and send it to the Backend and Database.
- **Interaction**:
  - Performs inference, retraining and sends results, performance metrics, and logs to the **Backend** for visualization and storage in the **Database**.

Database

- **Purpose**: Store pre-trained models, as well as logs, metrics, and retraining data.
- **Key Functions**:
  - Store the trained machine learning models and versioning.
  - Log model performance metrics, concept drift events, and retraining data.
  - Provide persistence for all relevant system data (training, testing, inference).
- **Interaction**:
  - Stores model data, results, performance metrics, and logs.

Backend Component

- **Purpose**: Acts as the orchestrator of the entire platform, managing communication between components, handling user requests, and preparing data for the Frontend.
- **Key Functions**:
  - Receive model results and metrics from the ML Component.
  - Communicate detected concept drift events and status updates from the Concept Drift Component.
  - Store logs, metrics, and model data in the Database.
  - Serve the Frontend with up-to-date information (visualizations, metrics, drift status).
- **Interaction**:
  - Orchestrates interactions and handles communication between all components, sends data to **Frontend**, and logs data in the **Database**

Frontend Component

- **Purpose**: Provides the interface where users interact with the platform, view model performance metrics, and see how models handle live data and concept drift.
- **Key Functions**:
    - Display real-time visualizations and metrics for the models' performance.
    - Show drift alerts and retraining status.
    - Enable the user to interact with different models and view data on demand.
- **Interaction**:
    - Displays real-time visualizations, metrics, and other information from the **Backend** for user interaction.

User Stories

- As a User, I want to see the estimated time of arrival predictions for various routes so that I can plan my travel more efficiently.
- As a User, I want to see the traffic flow predictions for my area so that I can avoid traffic congestion and save on fuel and time.
- As a User, I want to monitor and analyze parking availability to find optimal parking spots quickly and reduce my parking search time.
- As a User, I want to observe how the models handle concept drift due to significant changes like increased traffic or weather conditions so that I can trust their robustness in real-world scenarios.
- As a User, I want to receive notifications and visualizations of drift detection and adaptation activities so that I am aware of how the system is maintaining its performance over time.
- As a User, I want to use SUMO to simulate real-world traffic scenarios.
- As a User, I want an intuitive interface that displays real-time model performance and metrics, so that I can easily analyze and compare the accuracy of different predictive models.
- As a User, I want to test pre-trained models on live traffic simulations, so I can determine their robustness and fine-tune them for better accuracy in real-world scenarios.